

Etienne Lanzeray

etienne.lanzeray.etu@univ-lille.fr
+33 6 98 29 96 42
Lille, France

GitHub
Website

INTERESTS	Mathematics and Machine Learning. Planning to pursue a PhD in ML approximations of PDE solutions. I'm also interested in 3D rendering (Overview article).	
SKILLS	Data Science: Neural architectures for PDEs, PAC learnability and Rademacher bounds, Optimization, Bayesian Learning, NLP, Computer Vision, Reinforcement Learning, HPC, Fairness & Privacy. Mathematics: Distribution theory & PDEs, Spectral theory & Harmonic Analysis, Probabilities & Stochastic Processes, Statistics, Algebra, Analytic Number theory, Algebraic Topology & Differential Geometry, Complex Analysis. Programming: Python (Pandas, Scikit-learn, PyTorch, TensorFlow), Rust, SQL. Parallel & GPU: Open MPI, CUDA. Languages: French — native; English — TOEIC 970/990 (2025).	
EDUCATION	Data Science, Master <i>Université de Lille, Centrale Lille & IMT Nord Europe</i> <i>M1 with highest honors (2024–2025)</i> <i>M2 (2025–2026)</i>	<i>In english & Research Oriented</i> <i>Lille, France</i>
	Pure Mathematics, Master 1 <i>Université Paris Saclay (2017–2018)</i>	<i>Magistère Jacques Hadamard</i> <i>Orsay, France</i>
	Pure and Applied Mathematics, License <i>Université Paris Saclay (2016–2017)</i>	<i>Magistère Jacques Hadamard</i> <i>Orsay, France</i>
	Classes préparatoires, PCSI → PSI* → PSI* <i>Lycée Gustave Eiffel (2013–2016)</i>	<i>Bordeaux, France</i>
RESEARCH PROJECTS	Applying homogeneous Neural Networks to fluid mechanics models via homogeneous Galerkin projection Advisors: Andrey Polyakov (Inria & CRISAL, CNRS) Mihaly Petreczky (CRISAL, CNRS)	2025–2026
	Performance evaluation and generalization bounds of homogeneous Neural Networks Advisors: Andrey Polyakov (Inria & CRISAL, CNRS) Mihaly Petreczky (CRISAL, CNRS)	2024–2025 (Report)
	The analytic large sieve and applications to number theory Advisor: Etienne Fouvry (Institut de Mathématiques d'Orsay) (Report in french)	2017–2018
	Uniqueness of trigonometric series expansions Advisor: Julien Sabin (Institut de Mathématiques d'Orsay)	2016–2017

REFERENCES	Pierre Tirilly	Associate Professor at Université de Lille, Researcher at CRIStAL (CNRS), Head of the M1 Data Science program pierre.tirilly@univ-lille.fr
	Andrey Polyakov	Researcher at Inria Lille, Associate Researcher at CRIStAL (CNRS) andrey.polyakov@inria.fr
	Mihaly Petreczky	Researcher at CRIStAL (CNRS) mihaly.petreczky@centralelille.fr
RESEARCH EXPERIENCE	Research Intern	2025
	<i>Université de Lille</i>	
	Topic:	Performance evaluation and generalization bounds of homogeneous Neural Networks
	Advisors:	Andrey Polyakov (Inria & CRIStAL, CNRS) Mihaly Petreczky (CRIStAL, CNRS)
	Research Intern	2016
	<i>ICMCB (CNRS) & CEA Saclay</i>	
	Topic:	Mathematical modeling of the growth of nodules on silica nanoparticles and simulation implementation
	Advisor:	Etienne Duguet (ICMCB, CNRS)
TEACHING EXPERIENCE	Tutor in Computer Science (L1)	2024–2025
	<i>Université de Lille</i>	
	Topics:	Programming fundamentals, algorithms, basic data structures
	Description:	Support for first-year students during tutorials and labs
	Private Teacher	2019–2020
	<i>Acadomia</i>	
	Subjects:	Mathematics, physics–chemistry (secondary level)
	Oral Examiner (CPGE)	2017
	<i>Lycée Gustave Eiffel, Bordeaux</i>	
	Role:	Preparation of students for French engineering school entrance exams (30–45 min mathematics orals)
	Data Science Intern	2023–2024
	<i>Azur Drones, Mérignac</i>	
OTHER EXPERIENCE	Work:	Designed a flight database (TimescaleDB), built analytics for behavior and predictive maintenance, trained an LSTM autoencoder for anomaly detection on ~ 80 features ($> 90\%$ accuracy)
	Volunteer ML Collaborator	2023–2024
	<i>Ebiose (Inria Startup Studio)</i>	
	Project:	Contrastive-learning pipeline for selecting and scoring AI agents on <i>GSM8k</i>
	Repo:	github.com/EtiNL/Ebiose_CLGP
	Student Association Roles	2016–2018
	<i>Université Paris Saclay</i>	
	Roles:	Event manager for the Institut Mathématique d’Orsay (BDE Math $\sim$) Network and activities manager of the CROUS residence association (Eminet)
HOBBIES	Backpacking and multi-day high-mountain trekking, Skateboard, Chess & Guitar	